

(a) Adaptive / composite

OPEN GAP



Unknown and compound attacks

Generator drift interacts with entry and reference



TESTABLE EXPERIMENT



Joint held-out test

Withhold families and time; vary entry and privileges



REQUIRED EVIDENCE



Frozen-point evidence

Sensitivity, latency, legitimate completion and adaptation permissions

(b) Rare-event / inclusive

OPEN GAP



Rare events and access

Ranking omits review load and exclusion of users



TESTABLE EXPERIMENT



Threshold / fallback test

Vary prevalence scenarios, review and alternate routes



REQUIRED EVIDENCE



Burden, access and privacy

Intervals, subgroup/device outcomes, retention and sensitive exposure

(c) Workflow validity

OPEN GAP



Component-to-action gap

Module acceptance differs from unauthorized completion



TESTABLE EXPERIMENT



Binding ablation

Hold scores fixed; vary session and action binding



REQUIRED EVIDENCE



Versioned outcomes

Authorized tests, versions, privilege boundaries and final action outcomes

Research agenda: pair each gap with a falsifiable experiment and the evidence needed to interpret it.